EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) If the volume occupied by oxygen gas (O_2) at STP is 44.8dm^3, the number of molecules of O_2 in the vessels are:

(a) 3.01×10^{23} (b) 6.02×10^{23}
(c) 12.04×10^{23} (d) 24.08×10^{23}

(ii) The number of carbon atoms in 1 mole of sugar ($C_{12}H_{22}O_{11}$) are approximately:

(a) 6×10^{23} (b) 24×10^{23}
(b) (c) 60×10^{23} (d) 72×10^{23}

(iii) In the reaction $2Na + 2H_2O \rightarrow 2NaOH + H_2$, if 23g of Na reacts with excess of water, the volume of hydrogen gas (H_2) liberated at STP should be:

(a) 11.2dm^3 (b) 22.4dm^3 (c) 33.6dm^3 (d) 44.8dm^3

(iv) Which of the following sample of substances contains the same number of atoms as that of 20g calcium:

(a) 16g S (b) 20g C (c) 19g K (d) 24g Mg

(v) Which of the following statement is incorrect?

(a) The mass of 1 mole Cl_2 gas is 35.5g
(b) One mole of H_2 gas contains 6.02×10^{23} molecules of H_2
(c) Number of atoms in 23g Na and 24g Mg are equal
(d) One moles of O_2 at S.T.P occupy 22.4 dm^3 volume

(vi) For Avogadro's number, this statement is incorrect:

(a) It is the no. of particles in one moles of any substances
(b) Its numerical value is 6.02×10^{23}
(c) Its value change if temperature increases
(d) Its value change if number of moles increases

(vii) The minimum number of moles are present in:

(a) 1 dm^3 of methane gas at STP (b) 5 dm^3 of helium gas at STP
(c) 10 dm^3 of hydrogen gas at STP (d) 22.4 dm^3 of chlorine gas at STP

(viii) Number of atoms in 60g carbon are:

(a) 3.01×10^{23} (b) 3.01×10^{24}
(c) 6.02×10^{23} (d) 6.02×10^{24}

(ix) Maximum number of molecules present in the following sample of gas:

(a) 100g O_2 (b) 100g CH_4
(c) 100g CO_2 (d) 100g Cl_2

(x) Generally actual yield is:

(a) Greater than theoretical yield
(b) Less than theoretical yield
(c) Equal to the theoretical yield
(d) Some times greater and some times less than theoretical yield

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EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) Bohr's theory is not applicable to which of the following species.

(a) H (b) H^+

(c) He^{+1} (d) Li^{+2}

(ii) Nitrogen has the electronic configuration $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$ and not $1s^2 2s^2 2p_x^2 2p_y^1$

This is determined by.

(a) Aufbau principle (b) Pauli's rule

(c) Hund's rule (d) $n + \ell$ rule

(iii) Quantum number values for 3s orbital are

(a) $n = 0, \ell = 1$ (b) $n = 1, \ell = 0$

(c) $n = 3, \ell = 1$ (d) $n = 3, \ell = 0$

(iv) The radius of first orbit of hydrogen atom is …..

(a) 529 Å (b) 52.9 Å

(c) 5.29 Å (d) 0.529 Å

(v) Line spectrum is used as a tool for the identification of ….

(a) Colors (b) Electrons

(c) Elements (d) Molecules

(vi) In 1935 A.D. James Chadwick was awarded Nobel Prize because …

(a) He discovered proton

(b) He discovered neutron

(c) He determined the radius of hydrogen atom

(d) He gave the rules for electronic configuration

(vii) When 4d orbital is filled, the next electron enter into

(a) 5s (b) 5p

(c) 5d (d) 6s

(viii) Which of the following is not an iso electronic pair

(a) Na^+ and Ne (b) Na^+ and F^-

(c) Na and Ca (d) Na^+ and Mg^{+2}

(ix) Balmer series appears in the hydrogen spectrum if electron jump from any appropriate higher energy orbit to

(a) Second orbit (b) Third orbit

(c) Fourth orbit (d) Fifth orbit

(x) The shape of orbital for which $\ell = 0$ is

(a) Spherical (b) Dumbbell

(c) Double dumbbell (d) Complicated

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EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) If the bond angle in AB_2 type molecule is 104.5°, it geometry should be:
(a) Linear (b) Pyramidal (c) Bent (d) Planar Trigonal

(ii) The highest bond energy in the following is:
(a) Cl – Cl (b) H – F (c) H – O (d) H – N

(iii) The molecule which has zero dipole moment is:
(a) NH_3 (b) HCl (c) H_2O (d) CCl_4

(iv) The molecule which has maximum bond angle:
(a) CS_2 (b) H_2O (c) NH_3 (d) BF_3

(v) The shape and hybridization of BCl_3 molecule is:
(a) Tetrahedral and sp^3 (b) Linear and sp
(c) Planar trigonal and sp^2 (d) Angular and sp^3

(vi) Amongst the following molecules which one has trigonal pyramidal shape?
(a) SO_2 (b) CO_2 (c) NH_3 (d) C_2H_4

(vii) A simple covalent molecule possesses two bond pairs and two lone pairs around the central atom, its shape should be:
(a) Linear (b) Planar trigonal (c) Angular (d) Tetrahedral

(viii) The correct relation between Debye and coulomb meter is:
(a) $1D = 3.33 \times 10^{-30}Cm$ (b) $1D = 1.6 \times 10^{-19}Cm$
(c) $1D = 1.88 \times 10^{-12}Cm$ (d) $1D = 1.23 \times 10^{-8}Cm$

(ix) The bond order of N_2 molecule is:
(a) 0 (b) 1 (c) 2 (d) 3

(x) The number of sigma and pi bonds in C_2H_4 molecule are respectively:
(a) 3 and 1 (b) 2 and 2 (c) 5 and 1 (d) 4 and 2

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- **Real gas** is that which obey van der Waal equation. It reaches to ideal behavior at high temperature and low pressure.
- **Dalton law of partial pressure** tells that the total pressure of the mixture of two or more non reacting gases is equal to the sum of partial pressure of individual gases in a fixed volume enclosed container.
- **Graham's law of diffusion** tells that the rate of diffusion of a gas is inversely proportion to the square root of its density or molecular mass keeping the temperature and pressure constant.
- **Critical temperature** is the highest temperature at which a gas exists as liquid.
- **Critical pressure** is the pressure which is required at critical temperature to convert a gas into liquid.
- **Liquid air** Ordinary air on compression and cooling to extremely low temperature becomes liquefies and known as liquid air. It is mainly used for the cooling purpose.
- **Plasma** is the fourth state of matter. It is a mixture of ions and un ionized atoms and molecules.

EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) According to Graham's Law of diffusion, the ratio of diffusion of H_2 and O_2 are respectively:

(a) 1:2 (b) 2:1 (c) 1:4 (d) 4:1

(ii) Collection of gas over water is an example of:

(a) Graham's law (b) Dalton's law
(c) Avogadro's law (d) Gaylussac law

(iii) The molar volume of oxygen gas is maximum at:

(a) 0°C and 1 atm (b) 0°C and 2 atm
(c) 25°C and 1 atm (d) 25°C and 2 atm

(iv) The diffusion rate of C_3H_8 and CO_2 are same because:

(a) Both are poly atomic gases (b) Both are denser than air
(c) Both have same molar mass (d) Both contains carbon atoms

(v) The volume of gas would be theoretically zero at:

(a) 0°C (b) 0 K (c) 273 K (d) 273°C

(vi) Real gas reaches the ideal behavior at:

(a) Low temperature and low pressure (b) High temperature and high pressure
(c) Low temperature and high pressure (d) High temperature and low pressure

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(vii) Which one of the following statement is incorrect about the gas molecules?
(a) They have large spaces
(b) They possess kinetic energy
(c) Their collision is elastic
(d) Their molar mass depends upon temperature

(viii) If the Kelvin temperature of ideal gas is increase to double and pressure is reduce to one half, the volume of gas will:
(a) Remains same (b) Double
(c) Reduced to half (d) Four time

(ix) The molar volume of oxygen gas is 22.4 dm^3 at:
(a) 0°C and 1 atm (b) 25°C and 0.5 atm
(c) 0 K and 1 atm (d) 25 K and 0.5 atm

(x) Under similar condition CH_4 gas diffuses..... times faster than SO_2 gas:
(a) 1.5 time (b) 2 times
(c) 4 times (d) 16 times

Short Questions

1. State the following gas laws.
(i) Avogadro's law (ii) Charles law
2. State main postulates of kinetic molecular theory of gas.
3. Explain the following.
(i) Pressure and its various units (ii) Absolute zero
4. What is liquid air? Mention its three uses.
5. What is plasma? Give its significance in daily life.

Descriptive Questions

1. State and explain Dalton's law of partial pressure. Give practical applications of Dalton's law.
2. Derive general gas equation. Also deduce the value of R in atm dm^3/mol.K and J/mol.K.
3. How an ideal gas is differentiated from real gas? What are the causes of deviation of real gas from ideal behavior? Explain this deviation at low temperature and high pressure.
4. What is meant by diffusion and effusion? Explain Graham's law of diffusion.

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EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) The boiling point of water (H_2O) is 100°C where as that of hydrogen sulphide (H_2S) is 42°C, This can be attributed to:
(a) Smaller bond angle of H_2S than H_2O
(b) Smaller radii of oxygen than sulphur
(c) High I.P of oxygen than sulphur
(d) Tendency of water to form hydrogen bond

(ii) Liquids can form convex meniscus in a narrow glass tube when:
(a) Cohesive forces are stronger than adhesive forces
(b) Adhesive forces are stronger than cohesive forces
(c) Cohesive and adhesive forces are equal in strength
(d) None of these

(iii) Which of the following statement is incorrect?
(a) Viscosity is the resistance against flow of liquid
(b) Liquid possess definite volume
(c) One feels sense of cooling after bath due to condensation
(d) Sublimation is endothermic process.

(iv) Which of the following possesses weakest London dispersion forces:
(a) Cl_2 (b) F_2 (c) Br_2 (d) I_2

(v) Hydrogen bond is not found in:
(a) H_2O (b) CH_4 (c) NH_3 (d) HF

(vi) The boiling points of different liquids may be different at the same external pressure due to:
(a) Amounts of liquids (b) Intermolecular forces
(c) Surface area (d) Viscosities

(vii) Which statement is incorrect about evaporation?
(a) It is an exothermic process
(b) It is a reverse process of condensation
(c) It occurs at all temperature and pressure
(d) It causes cooling effect

(viii) Which of the following liquids show maximum surface tension?
(a) Water (b) Mercury (c) Ethyl alcohol (d) Gasoline

(ix) (ix) A non-polar molecule with bigger size will experience:
(a) London forces (b) dipole-dipole interaction
(c) Hydrogen bonding (d) All of these

(x) Cooking time is reduced in a pressure cooker because:
(a) Boiling point of water increases (b) Boiling point of water decreases
(c) Vapor pressure of liquid is reduced (d) Heat is uniformly distributed

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EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) Quantity of energy released when 1 mole of the ionic crystal is formed from the gaseous ions is called:

(a) Bond energy (b) Potential energy
(c) Ionization energy (d) Lattice energy

(ii) Graphite can conduct electricity in one direction only.With this behavior, it is said to be:

(a) Amorphous (b) Symmetrical
(c) Anisotropic (d) Polymorphous

(iii) NaCl and MgO are isomorphous solids because these have same:

(a) Melting point (b) Boiling point
(c) Geometrical shapes (d) Chemical Properties

(iv) The temperature at which two allotropic forms co-exist in equilibrium is called as:

(a) Melting temperature (b) Fusion temperature
(c) Transition temperature (d) Critical temperature

(v) Diamond is the example of:

(a) Ionic solid (b) Covalent solid
(c) Metallic solid (d) Molecular solid

(vi) Ice is a molecular solid, the intermolecular force of attraction among its molecules are:

(a) Ionic bond (b) Covalent bond
(c) Metallic bond (d) Hydrogen bond

(vii) If all three axes in unit cell are of different length but all angles are of 90° then it is identified as:

(a) Orthorhombic system (b) Cubic system
(c) Tetragonal system (d) Hexagonal system

(viii) In NaCl, each Na^+ ion is surrounded by Cl^- ions in the numbers:

(a) Four (b) Three
(c) Six (d)Seven

(ix) It is an example of crystalline solid:

(a) Alum (b) Glass
(c) Plastic (d) Rubber

(x) Each unit cell of sodium chloride consists of:

(a) Eight Na^+ and Eight Cl^- ions (b) Two Na^+ and Two Cl^- ions
(c) Four Na^+ and Four Cl^- ions (d) Four Na^+ and Six Cl^- ions

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EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) In the equilibrium system of $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$, the correct relationship between Kc and Kp is:

(a) Kp> Kc (b) Kp< Kc (c) Kp = Kc (d) $\frac{K_p}{K_c} = 1$

(ii) The equilibrium of which of the following reaction would not be affected by an increase in pressure:

(a) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ (b) $2NO_{(g)} + Cl_{2(g)} \rightleftharpoons 2NOCl_{(g)}$

(c) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ (d) $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$

(iii) What happens to the value of Kc when a catalyst is added to a chemical system at equilibrium?

(a) It decreases (b) It increases

(c) It becomes zero (d) It remains unchanged

(iv) The correct Ksp expression of a sparingly soluble salt $Li_2C_2O_4$ among the following is:

(a) $Ksp=[Li^+][C_2O_4^{-2}]$ (b) $Ksp=[Li^+]^2[C_2O_4^{-2}]$

(c) $Ksp=[2Li^+][C_2O_4^{-2}]$ (d) $Ksp=[2Li^+]^2[C_2O_4^{-2}]^2$

(v) If the equilibrium expression of a reversible reaction is

$$Kc = \frac{[C]^2}{[A][B]}$$

The balanced equilibrium equation should be:

(a) $C \rightleftharpoons A + B$ (b) $A + B \rightleftharpoons C$

(c) $2C \rightleftharpoons A + B$ (d) $A + B \rightleftharpoons 2C$

(vi) The equilibrium of $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ (ΔH = +ve) is affected by change in:

(a) Temperature only (b) Pressure only

(c) Both temperature and pressure (d) Neither temperature nor pressure

(vii) In the reaction $A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)}$, the equilibrium constants Kp = Kc because:

(a) $\Delta n > 1$ (b) $\Delta n < 1$ (c) $\Delta n = 1$ (d) $\Delta n = 0$

(viii) The term active mass use in law of mass action means:

(a) No. of mole (b) No. of molecules

(c) mole per dm^3 (d) gram per dm^3

(ix) NaCl when added to an aqueous silver chloride solution:

(a) Decreases the solubility of AgCl (b) Increases the solubility of AgCl

(c) Forms a clear solution (d) Does not effect

(x) Some reactions are nearly to completion in the forward direction and identified by their:

(a) Very high value of Kc (b) Very low value of Kc

(c) Very high value of ΔH (d) Very low value of ΔH

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EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) H_2SO_4 is stronger acid than CH_3COOH because:

(a) It gives two H^+ ion per molecule (b) Its boiling point is high

(c) Its degree of ionization is high (d) It is highly corrosive

(ii) Al_2O_3 is amphoteric oxide because it reacts with:

(a) Acids (b) Base

(c) Both acids and base (d) neither acid nor base

(iii) Which of the following is not a Buffer solution:

(a) CH_3COOH/CH_3COONa (b) $Na_2CO_3/NaHCO_3$

(c) NH_4OH/NH_4Cl (d) NaOH/HCl

(iv) Which oxide is amphoteric in nature:

(a) K_2O (b) CO_2

(c) CaO (d) Al_2O_3

(v) Which of the following does not alter the pH of a solution?

(a) NH_4Cl (b) Na_2CO_3

(c) NaCl (d) Mg(OH)Cl

(vi) Conjugated acid of NH_3 is:

(a) NH_4^+ (b) NH_2^-

(c) NH_2 (d) NH

(vii) Salt which is formed by the neutralization of weak acid and strong base is:

(a) $NaNO_3$ (b) NH_4Cl

(c) Na_2CO_3 (d) NH_4CN

(viii) Which of the following statements is not correct about the bases?

(a) They have bitter tastes (b) They have high pH value

(c)They react with acids to form salts (d) They turn blue litmus red

(ix) A conjugate acid base pair has the difference of only:

(a) One electron (b) One proton

(c) One electron pair (d) One proton pair

(x) Salt formed by neutralization of weak acid and weak base is:

(a) NH_4Cl (b) Na_2CO_3

(c) NH_4CN (d) Na_2SO_4

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EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) For a hypothetical reaction x + y → z, if the conc. of x is double, the rate increases by square and if the conc. of y is doubled the rate increases by twice. The experimental rate law of this reaction is:

(a) $R = K [x]^1 [y]^1$ (b) $R = K [x]^1 [y]^2$
(c) $R = K [x]^2 [y]^1$ (d) $R = K [x]^2 [y]^2$

(ii) The unit of rate constant (K) for the first order reaction is:

(a) s^{-1} (b) conc. s^{-1}
(c) $conc^{-1}$. s (d) $conc.^{-1} s^{-1}$

(iii) Rate constant of a reaction is affected by:

(a)Conc. of reactants (b) Conc. of products
(c) Temperature (d) Reaction time

(iv) The purpose of using of catalyst in a chemical reaction is to change:

(a) Equilibrium constant (b) Enthalpy of reaction
(c) Activation energy (d) Nature of reaction

(v) The overall order of reaction to which the rate law is R = K:

(a) Zero order (b) First order
(c) Second order (d) Third order

(vi) Ionic reactions are classified into:

(a) Slow reaction (b) Moderate reactions
(c) Fast reactions (d) Reversible reaction

(vii) The decomposition of H_2O_2 is inhibited by:

(a) 2% ethanol (b) Glycerin
(c) MnO_2 (d) V_2O_5

(viii) The best alternative term for the velocity of reaction is:

(a) Rate of appearance (b) Rate of disappearance
(c) Average rate (d) Instantaneous rate

(ix) Order of reaction is the power to which concentration of reactant is:

(a) Lowered (b) Raised
(c) Stopped (d) Constant

(x) In the reaction A → B, the rate of disappearance is written as:

(a) $\frac{dA}{dt}$ (b) $\frac{-dA}{dt}$
(c) $\frac{dB}{dt}$ (d) $\frac{-dB}{dt}$

Short Questions

1. Define the following:

(a) Rate of reaction (b) Velocity of reaction
(c) Order of reaction (d) Rate constant

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EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) Only one pair of liquid in the following set does not obey Roault law, identify it:

(a) Methanol and Ethanol (b) Benzene and toluene
(c) n-Hexane and n-heptane (d) Ethanol and Acetone

(ii) Identify the incorrect statement about colloidal solution:

(a) It showsTyndall effect (b) Its particles movement is Brownian type
(c) Its particle size is less than 1 nm (d) Its physical appearance is translucent

(iii) Effect of pressure change play significant role in the solubility of:

(a) Solid into liquid (b) Liquid into liquid
(c) Gas into liquid (d) All of them

(iv) Which is not a colligative property:

(a) Lowering in vapours pressure (b) Elevation in boiling point
(c) Depression in freezing point (d) Atmospheric pressure

(v) According to Raoult law the relative lowering of vapour pressure is equal to:

(a) Mole fraction of solvent (b) Mole fraction of solute
(c) Molality (d) Molarity

(vi) The sum of mole fractions of components of a solution is equal to:

(a) 0.0 (b) 1.0
(c) 10 (d) 100

(vii) How many moles of NaOH are present in a $2dm^3$ of 1 molar aqueous solution of it.

(a) 0.5 mole (b) 1 mole
(c) 1.5 mole (d) 2 mole

(viii) A colloidal solution of liquid into liquid is known as:

(a) Gel (b) Foam
(c) Sol (d) Emulsion

(ix) An example of completely immiscible liquid pair is:

(a) Benzene to toluene (b) Water and phenol
(c) Water and Benzene (d) Water and methanol

(x) A 15% W/W KOH solution can be prepared by mixing 15g KOH in:

(a) 15g water (b) 85g water
(c) 100g water (d) 115g water

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- **Standard Enthalpy of Reaction, ΔH°** is the heat change of a chemical reaction and can be measured by subtracting the sum up values of ΔH_f° of products to the sum up values of ΔH_f° of reactants.
- **Hess's Law** tells that if a reaction is carried out through several steps then the enthalpy change for the overall reaction must be equal to the sum of enthalpy change of these individual steps.
- **Born Haber Cycle** is an application of Hess's Law and generally use to determine the Lattice of ionic compounds.

EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) Least entropy found in which of the following state of water:

(a) Liquid water at 4°C (b) Liquid water at 25°C
(c) Steam at 100°C (d) Ice at 0°C

(ii) The energy corresponds to the given thermochemical process is labeled as:

$$Li^{+}_{(g)} + \bar{C}l_{(g)} \longrightarrow LiCl_{(s)}$$

(a) Ionization energy (b) Lattice energy
(c) Enthalpy of combustion (d) Enthalpy of atomization

(iii) Which of the following change is not an endothermic reaction:

(a) Decomposition of lime (b) Cracking of alkanes
(c) Combustion of butane (d) Photosynthesis

(iv) Heat transfer can not be feasible across the boundryof a:

(a) Open system (b) Close system
(c)Isolated system (d) Thermo permeable system

(v) Which of the following enthalpy change is always negative:

(a) Enthalpy of formation (b) Enthalpy of decomposition
(c) Enthalpy of combustion (d) Enthalpy of reaction

(vi) Standard enthalpy of formation of all of the following elements at 25°C and 1 atm pressure are zero except:

(a) $C_{(diamond)}$ (b) $C_{(graphite)}$
(c) O_2 (d) N_2

(vii) In the equation of First law of thermodynamics (ΔE = q + w), the property(s) which depends upon initial and final state is (are):

(a) ΔE (b) q
(c) W (d) Both q and W

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(viii) Which statement is incorrect:
(a) For constant pressure process, $\Delta H = \Delta E + P\Delta V$
(b) For constant volume process, $\Delta E = q$
(c) For exothermic reactions, $\Delta H \rangle 0$
(d) For Hess law $\sum \Delta H^\circ_{(cycle)} = 0$

(ix) In a thermochemical process, no work is done if the system is kept at:
(a) Constant temperature (b) Constant pressure
(c) Constant volume (d) Constant mass

(x) Volume is a:
(a) State function (b) Path function
(c) Intensive properties (d) Colligative property

Short Questions

1. State precisely the meaning of each of the following terms.
(i) System and surrounding (ii) State and state function
(iii) Internal energy and Enthalpy
2. What is meant by Internal energy change (ΔE) and Enthalpy change (ΔH)? Under what conditions are ΔE and ΔH equal?
3. How can you define standard enthalpy of formation and standard enthalpy of reaction.

Descriptive Questions

1. State and explain First law of Thermodynamics. Derive pressure-volume work of a system.
2. Discuss the applications of the First law of thermodynamics at constant pressure and constant volume.
3. State and explain Hess's Law of enthalpy summation. Discuss its applications.
4. Explain Exothermic and Endothermic reactions with the help of the energy diagram.

Numerical Questions

1. A thermochemical process is carried out at constant pressure of 8.52atm. If it absorbs 15.4KJ energy from the surrounding due to which an expansion in the volume of 4.7dm^3 is occurred. Calculate its change in internal energy. [Ans: 11.34KJ]
2. Using the data in table 11.2, calculate the standard enthalpy change for each of the following reactions.

$$2H_2S_{(g)} + 3O_{2(g)} \longrightarrow 2H_2O_{(\ell)} + 2SO_{2(g)}$$
$$Fe_2O_{3(s)} + 3CO_{(g)} \longrightarrow 2Fe_{(s)} + 3CO_{2(g)}$$

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- **Lead storage battery** is a portable secondary battery use in automobile in which lead serves as anode and lead oxide (PbO_2) works as cathode.
- **Fuel Cell** is a special type of Galvanic cell which generates electricity by chemical reactions but a continuous supply of reactants and a continuous removal of product are required.

EXERCISE

Multiple Choice Questions

1. Choose the correct answer

(i) The outer body of dry cell serves as anode, it is made up of:
(a) Copper (b) Zinc
(c) Lead (d) Iron

(ii) The conduction of electricity through an electrolytic solution is due to the flow of:
(a) Electrons (b) Ions
(c) Atoms (d) Molecules

(iii) During electrolysis, the reaction that takes place at anode is:
(a) Oxidation (b) Reduction
(c) Hydrolysis (d) Simultaneous oxidation and reduction

(iv) This statement is not correct for lead storage battery:
(a) It can be recharged (b) It is a primary battery
(c) Anode is made up of lead (d) Cathode is made up of lead oxide

(v) In Zn-SHE voltaic cell, the half reaction occurs at anode is:
(a) $Zn^{+2} + 2\bar{e} \longrightarrow Zn$ (b) $Zn \longrightarrow Zn^{+2} + 2\bar{e}$
(c) $2H^{+} + 2\bar{e} \longrightarrow H_2$ (d) $H_2 \longrightarrow 2H^{+} + 2\bar{e}$

(vi) The strongest oxidizing agent in the electro chemical series is:
(a) Li (b) H_2
(c) Cu (d) F_2

(vii) Galvanized rode of iron is coated with:
(a) Nickel (b) Zinc
(c) Chromium (d) Carbon

(viii) Fuel cell is a typical Galvanic cell which is based on the reaction between:
(a) Nitrogen and oxygen (b) Hydrogen and oxygen
(c) Methane and oxygen (d) Hydrogen and zinc

(ix) Oxidation number of Cr in $Na_2Cr_2O_7$ is:
(a) + 3 (b) + 6
(c) + 8 (d) + 12

(x) Which of the following half cell reaction show oxidation:
(a) $Fe^{+3} \rightarrow Fe^{+2}$ (b) $Cl_2 \rightarrow 2Cl^{-}$
(c) $SO_4^{-2} \rightarrow SO_4^{-3}$ (d) $Zn \rightarrow Zn^{+2}$

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MCQS ANSWER KEYS

Chap #	MCQs No.									
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
1	c	d	a	a	a	c	a	b	b	b
2	b	c	d	d	c	b	b	c	a	a
3	c	b	d	a	c	c	c	a	d	c
4	d	b	c	c	b	d	d	d	a	c
5	d	a	c	b	d	b	a	b	a	a
6	d	c	c	c	b	d	a	c	a	c
7	a	c	d	b	d	a	d	c	a	a
8	c	c	d	d	c	a	c	d	b	c
9	c	a	c	c	a	c	b	d	b	b
10	d	c	c	d	b	b	d	d	c	b
11	d	b	c	c	c	a	a	c	c	a
12	b	b	a	b	b	d	b	b	b	d

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